



NILUS LAB

Reversing Cardiac Ageing Without Oncogene Risk

Generative cellular reprogramming therapeutics targeting cardiomyopathy.
Powered by the Zenith v28.0 GOLD single-cell foundation model.

LONGEVITY BIOTECH FELLOWSHIP (LBF) - ADMISSION INTERVIEW PRESENTATION

CLINICAL RISK & LIMITS

Cardiac Senescence & Reprogramming Limits

- **64M+ Heart Failure Patients**
Epigenetic decay and loss of contractility drive global mortality.
- **Oncogenic Risk of OSKM Factors**
Classic Yamanaka factors trigger uncontrolled dedifferentiation.
- **Delivery & Specificity Barriers**
Current therapies lack safe, cardiac-specific lineage targets.

ZENITH PLATFORM SOLUTION

Safe, Target-Specific Rejuvenation

- **Causal GCN Target Auditing**
PyTorch Causal GCN blocks oncogenic MYC loops and infers safe target links.
- **Trained on 500,000 Single Cells**
Maps cardiac manifolds via scVI variational autoencoders.
- **-11.9y Epigenetic Age Reversal**
Trained on real donor ages with a cross-validated MAE of 6.0y.

1 scVI Generative Core

- In-Silico Modeling Core
- Trained on 500,000 cells
- 4,908 variable gene inputs
- 20-dimensional latent VAE
- HCA + PERIHEART data

2 Causal GCN Safety Gate

- Gaussian-Kernel Autoencoder
- Causal Feature Reconstruction
- Laplacian-normalized GCN
- Blocks MYC / SNAIL pathways
- Dynamic target link prediction

3 AF3 & ElasticNet Clock

- AlphaFold 3 Valency models
- Models 3D TF-DNA docking
- ElasticNet age clock
- MAE = 6.0y on real HCA
- Transient dose windows

MARKET POSITIONING & COMPETITION

Target Market, Competition & Differentiation

- **CVD Longevity Market** — \$50B addressable cardiomyopathy indication targets.
- **Fellowship Collaboration** — Synergize with stealth replacement, tissue engineering, and AI founders.
- **Altos Bio & Retro Labs** — Broad systemic vectors. Nilus focuses on cardiac specificity.
- **Robotic Wet-Lab Loop** — Zenith bridges computational

WETLAB VALIDATION FUNDING ASK

Raising \$500,000 Seed round specifically to fund cocktail synthesis, calcium imaging, ATAC-seq chromatin profiling, and rodent safety assays.

\$500K Ask

WETLAB FUNDING ASK

-11.9 Years

EPIGENETIC AGE RESET (MAE=6.0y)

THE TEAM

Team Chemistry & Dynamics

- **Computational VAE Core:**
Expertise in deep latent manifolds.
- **Replacement Synergies:**
Aligning GCN predictions with organ replacement and robotics.
- **Fellowship Engagement:**
Partnering with LBF community to accelerate target translation.
- **BIO 2026 Summit Target:**
Preparing to showcase wetlab data.

DEVELOPMENT ROADMAP

Key Execution Milestones



Phase 1: Cocktail Synthesis & Wetlab Setup

Synthesize target cocktails and setup primary cardiomyocyte mod



Phase 2: Preclinical In-Vitro Assaying

Calcium-transient imaging & chromatin accessibility (ATAC-seq) p



Phase 3: In-Vivo Rodent Assays & Seed Close

Conclude rodent cardiomyopathy trials and close \$500,000 seed